Class Notes: Body Symmetry, Germ Layers and Coelom

Animal Kingdom • PU-I Biology • Class Notes

Body Symmetry

radial symmetry: any plane through the central axis divides the body into identical halves (e.g. coelenterates, ctenophores, echinoderms). Bilateral symmetry: only one plane divides the body into identical left and right halves (e.g. insects, chordates).

Germ Layers

Asymmetry: no plane divides the body into equal halves (e.g. sponges/Porifera). Diploblastic animals have ectoderm and endoderm with mesoglea in between (e.g. coelenterates); triploblastic animals have ectoderm, mesoderm and endoderm (e.g. Platyhelminthes to Chordates). Coelomates possess a mesoderm-lined body cavity (coelom) — e.g. Annelida, Mollusca, Arthropoda, Echinodermata, Hemichordata, Chordata.

Body Cavity

Pseudocoelomates have mesoderm as scattered pouches between ectoderm and endoderm — e.g. Aschelminthes. Acoelomates lack a body cavity altogether — e.g. Platyhelminthes.